Built on Midnight · live on Preview

Prove the lot.
Keep the measurements.

A parts supplier proves to its customer that a lot is inside the drawing tolerance, without handing over the measurements.

lotproof.avanc3.pe

github.com/kasbsquall/lotproof

The problem

Before accepting a lot, customers ask for every raw measurement. Suppliers know where those numbers end up.

> "Contract manufacturers are leery of sharing information that could lead to demands for more price cuts."
>
> Matt Littlefield, LNS Research, quoted by SAP

> Dimensional reports "can reveal details about the precision and consistency of a supplier's manufacturing process."
>
> A Peruvian manufacturer, anonymous (translated)

Today there are two options: send everything, or trust the certificate. Lotproof is a third one.

How it works

One contract, two parties, five moves.

1



The customer sets the rules

Publishes the drawing spec and approves one measuring station for it.

`publishSpec · registerStation`

2



The station seals the lot

A hiding commitment to the 32 values, bound to this lot and this drawing.

`commitLot`

3



The supplier proves the count

A zero-knowledge proof of how many parts sit inside the band. Changed numbers are refused.

`proveLot`

4



The customer asks again, later

A tighter band, answered from the same seal. No new measurement.

`requestAudit · answerAudit`

5



One part, if disputed

The supplier opens a single value of a proven lot. The other 31 stay private.

`disclosePart`

What is public and what stays private

The customer sees the verdict. The supplier keeps the process.

Public on Midnight	Private to the supplier
Drawing spec: nominal, band, acceptance number	The 32 measurements of each lot
The station's commitment, at measuring time	The randomness behind each commitment
Count inside the band and the verdict	Which parts are out of band
Lots committed, proven, audits requested and answered	Drawing numbers and lot labels (keyed hashes on chain)
Any part the supplier chooses to disclose	Every part it does not disclose

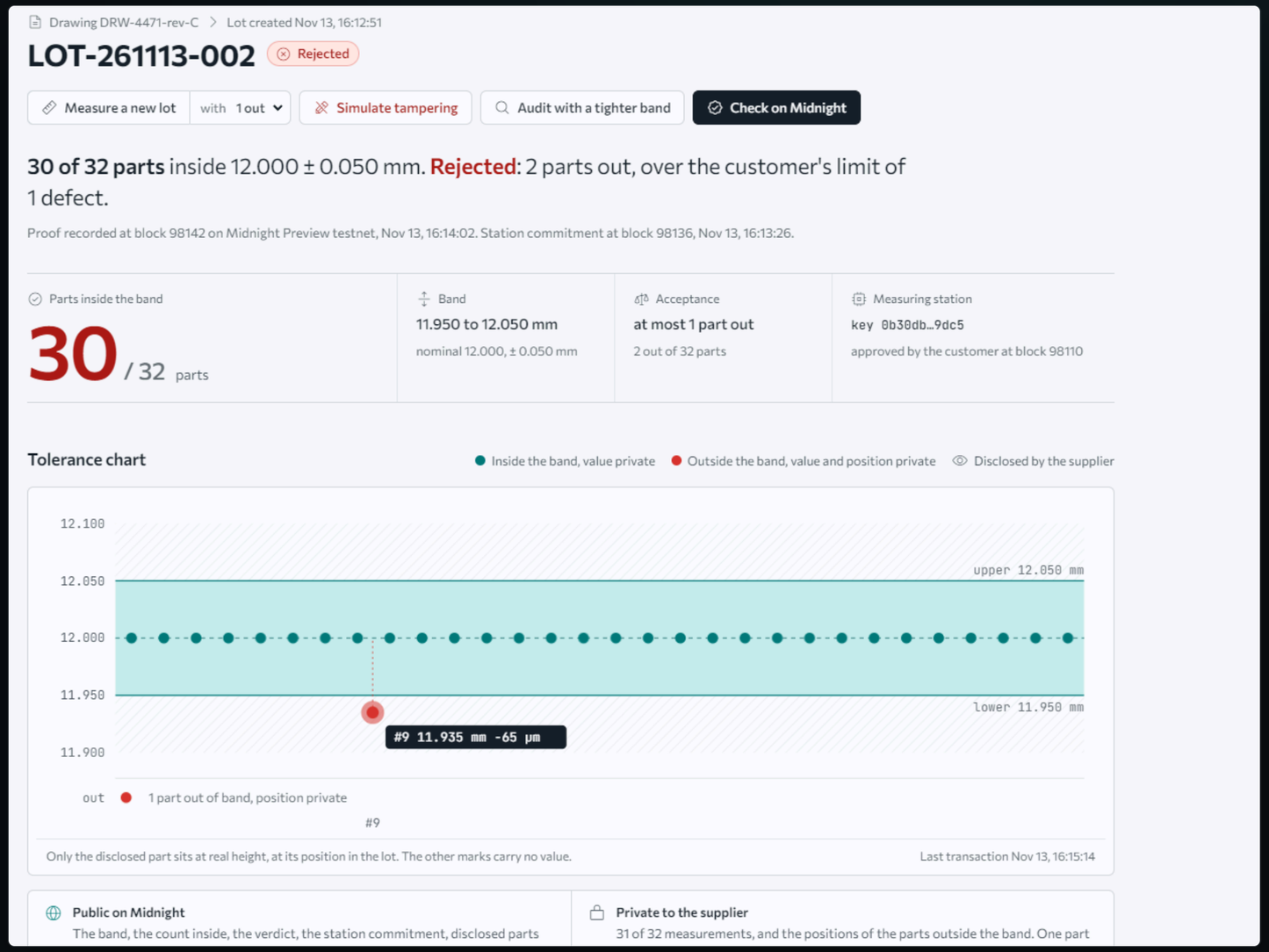
LOT-261113-002

30 /32

inside 12.000 ± 0.050 mm. Rejected.

The customer saw the count and the verdict. Not one diameter. Part #9 was disclosed at 11.935 mm; the other 31 stay private.

`commitLot block 98136 · proveLot block 98142 · discloseLot block 98154`



Why Midnight and not a signed report

A signed report only answers the questions asked on the day.

requestAudit · 12.000 ± 0.020 mm

17 /32

LOT-261113-001, accepted at 31 of 32, audited later against a tighter band. Proven from the commitment made at measuring time.

requestAudit block 98159 · answerAudit block 98167

- **A later question, without the data.** A field failure or a tightened tolerance gets an answer from the sealed values.
- **A disputed part, without the rest.** One value opens, proven to belong to the sealed lot.
- **No quiet selection.** Commitments, proofs and audits are counted in public, so an unproven lot or an unanswered question shows.

Compact 0.36 · Midnight.js 5 · proof server 9.0.0

Every claim is enforced in the contract and tested.

8

circuits

constructor, publishSpec, registerStation, revokeStation, commitLot, proveLot, disclosePart, requestAudit, answerAudit

32

simulator tests, in CI

17 of them are inputs the contract must refuse: changed values, reused commitments, wrong party, unknown lot or spec.

5

end-to-end tests on Preview

Real proofs, 12 transactions: accepted lot, rejected lot, tampering refused, disclosure, audit. Passed on 13 Nov 2026.

🚫 Tampering, live: `"Measurements do not match the station commitment"`. No count reaches the ledger.

We showed the demo to a manufacturer that buys from outside suppliers

A Peruvian manufacturer, anonymous, answered six questions.

- **Today:** certificates of conformity and dimensional reports as PDF or Excel, mostly for critical parts.
- **The pain:** suppliers do not all share complete measurements, and later "it can be hard to verify that the documents really belong to the lot that was delivered".
- **Who pays:** the supplier at first, per lot or monthly. A shared cost if the buyer saves inspection time. The customer verifies for free.
- **A real trial needs:** measurements from real production, independent verification, and integration with the instruments the plant already uses.

One buyer's view after seeing the demo. Not a pilot and not a purchase commitment.

Answers translated from Spanish

Everything in the repository is new in this wave

Done, next, and what Lotproof does not change.

Done

- ✓ Contract v3 on Preview with customer audits
- ✓ Public demo, no wallet needed
- ✓ Tests in CI and end to end on Preview
- ✓ Feedback from a real buyer

Next

- → A real lot from the manufacturer, run on Preview
- → Station adapter: commit the report the measuring software already exports
- → Bind the station key to the instrument

Limits, declared

- ⓘ If the instrument measures wrong, the proof is correct about wrong numbers
- ⓘ The station still sits in the supplier's plant

Team

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Ask

- **One supplier and buyer pair** to run a real lot through Lotproof.
- **An introduction to measuring software vendors** for the station adapter.
- **A place in Midnight Build Club** to take it from Preview to production.

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